

# Machine Learning Portfolio Construction & Risk Management

Professional portfolio case study based on a master thesis project

**Riccardo Cornelli**

MSc Economics & Finance | Quantitative asset management project

## What the project demonstrates

This portfolio summarizes a complete investment research pipeline: data engineering, supervised learning, walk-forward validation, stock ranking, portfolio construction, transaction cost analysis, Markowitz allocation, Monte Carlo simulation and stress testing. The objective is not to show isolated charts, but to make the full decision chain understandable from an asset-management and risk-management perspective.

**S&P 500**

investment  
universe

**17**

technical features

**Top 20%** **2014-2025**

monthly stock  
selection

out-of-sample  
backtest

## Core investment logic

Machine learning is used as a ranking engine, not as a perfect price forecaster. Forecasts are converted into cross-sectional scores; only the highest-ranked stocks are selected. The allocation layer then evaluates whether the selected universe can generate a superior risk-return profile versus passive benchmarks.

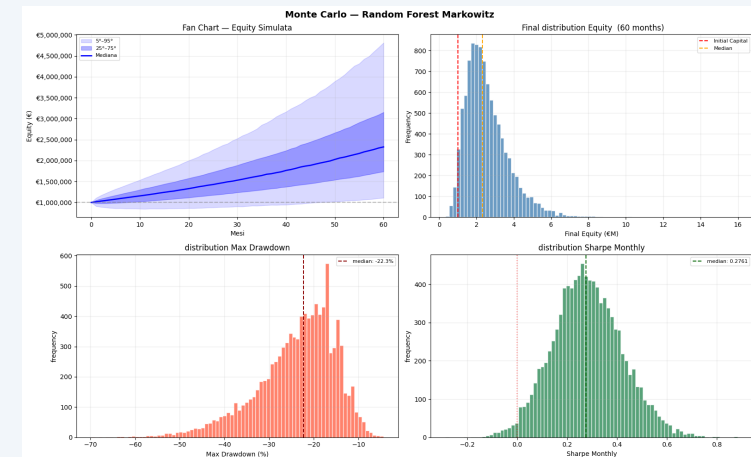


Exhibit: Monte Carlo distribution and path risk

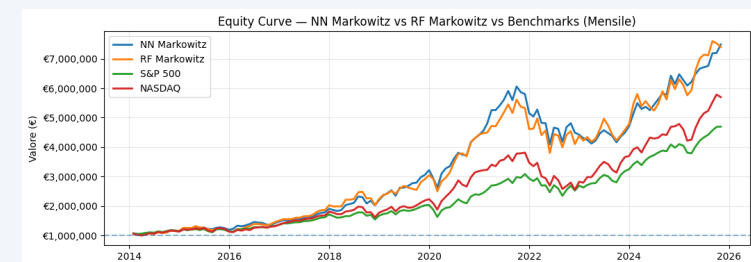


Exhibit: optimized portfolios and benchmarks

The empirical pipeline starts from the S&P 500 universe and builds a clean master long panel for model training and portfolio simulation.

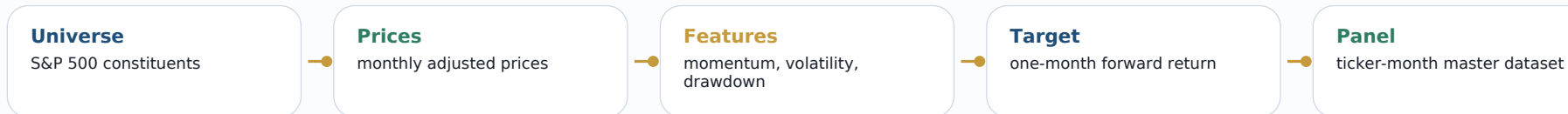
[illegible]

Exhibit: sample of stock-level monthly features

Historical price series were retrieved through R on a monthly frequency, using adjusted prices to reduce distortions from corporate actions.

At month  $t$ , each feature uses only information observable up to that point; the target is the realized return in month  $t+1$ .

The panel was cleaned for missing values, abnormal observations and discontinued histories. Assets enter only after sufficient data history.

The feature set includes momentum/trend indicators, multi-window volatility and maximum drawdown measures for downside and path-dependent risk.

- Using a ticker-month panel makes the process repeatable and auditable: every prediction can be traced back to the information available at that date.
- Monthly frequency reduces noise versus daily data and matches the investment horizon used for portfolio rebalancing.
- Drawdown and volatility features prevent the model from ranking stocks only by past growth; they add a risk-sensitive dimension to stock selection.

## 2. Machine learning layer: prediction as an asset-ranking signal

Random Forest and Neural Network models are embedded in the same monthly walk-forward protocol to avoid look-ahead bias and keep results comparable.

### Random Forest

Tree-based ensemble used to capture nonlinear relationships and interactions between trend, volatility and drawdown variables. Selected because it delivered the strongest cross-sectional IC profile.

### Neural Network

MLP regression architecture with imputation, standardization, ReLU hidden layers and early stopping. Used as a flexible nonlinear scoring model for complex return patterns.

#### Model comparison: error metrics vs ranking metrics

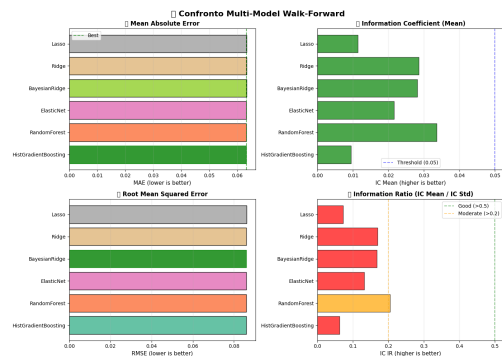


Exhibit: multi-model walk-forward comparison

#### Feature importance: which signals drive the ranking

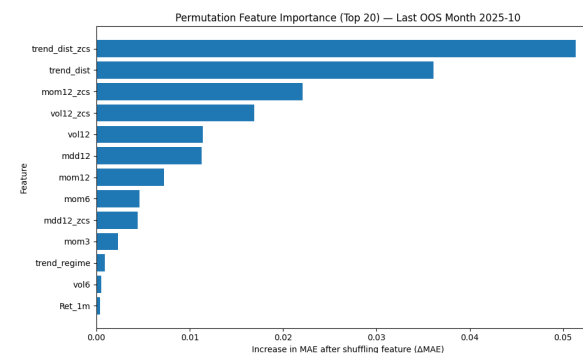


Exhibit: permutation importance on the final OOS month

**9,967**

OOS forecasts per tested model

**0.0337**

RF mean Information Coefficient

**0.204**

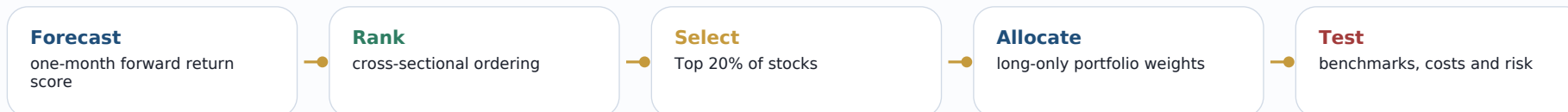
RF IC Information Ratio

**98k+**

NN out-of-sample predictions

# 3. From forecasts to investable portfolios

The models do not mechanically allocate capital to all stocks. They generate monthly scores, rank the cross-section, and select the highest-ranked names.



## Operating rule

- At each month, all investable stocks receive a predictive score from the model.
- The portfolio keeps only the top 20% of ranked names; the rest of the universe is excluded.
- The portfolio is long-only and rebalanced monthly; transaction costs are modeled at 0.15% one-way.
- Benchmarks include the S&P 500 and NASDAQ to separate true stock-selection alpha from broad market exposure.

## Predictive interpretation

The models are weak point forecasters, which is normal in equity-return prediction. Their value comes from ranking power: even a modest, persistent information signal can become economically useful once translated into a disciplined portfolio rule.

## Prediction vs realized returns: weak point forecasts, useful ranking logic

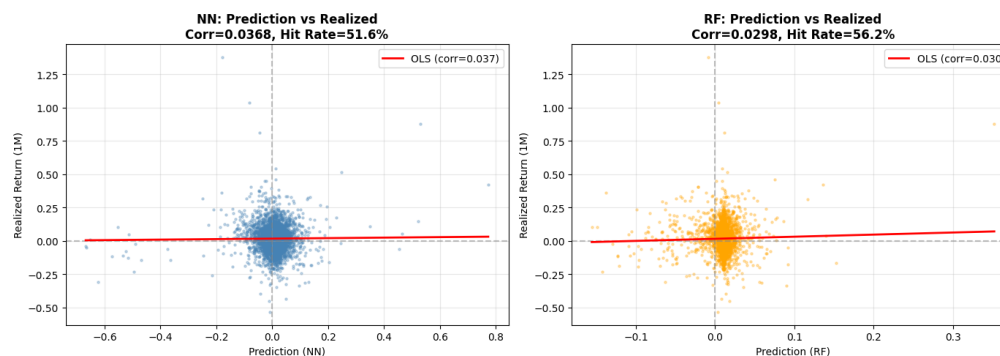


Exhibit: prediction-realization scatter for NN and RF

**0.0368**

NN  
prediction-realization  
correlation

**0.0298**

RF  
prediction-realization  
correlation

**56.2%**

RF upper-tail hit  
rate

**108/213**

NN positive-alpha  
months

# 4. Backtest evidence: growth, drawdown and benchmark discipline

The strategies are evaluated against passive benchmarks to understand whether the ML ranking adds value beyond simple market exposure.

## Performance and risk comparison across strategies

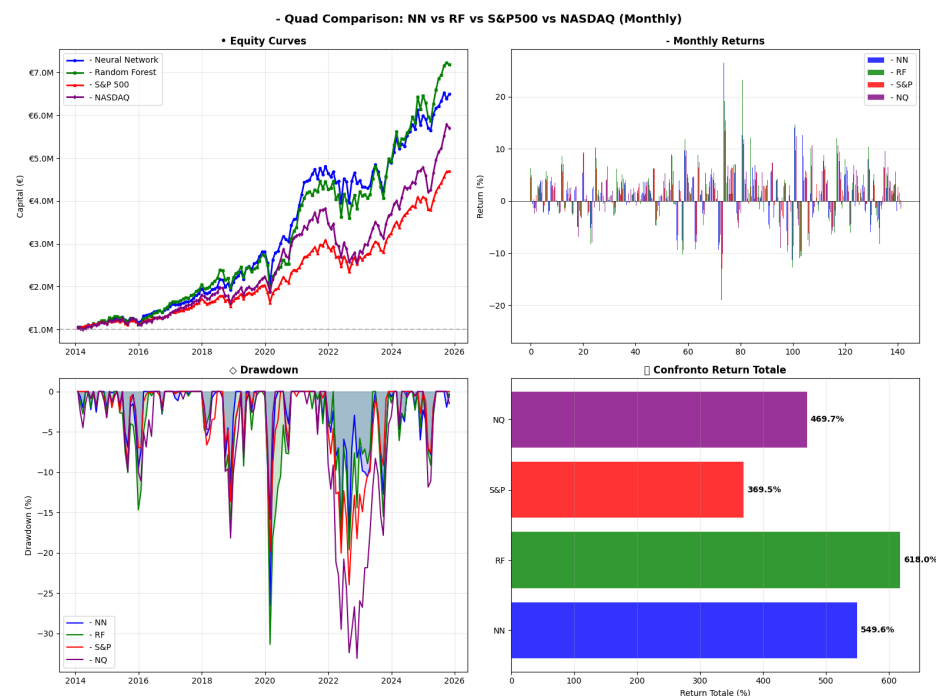


Exhibit: equity curve, returns, drawdown and performance bars

## Backtest summary

Common sample: Jan 2014 - Oct 2025, monthly rebalancing

| Strategy   | Total return | Volatility | Max DD  | Sharpe |
|------------|--------------|------------|---------|--------|
| NN Top 20% | 549.57%      | 18.37%     | -26.53% | 0.93   |
| RF Top 20% | 618.04%      | 20.59%     | -31.36% | 0.88   |
| S&P 500    | 369.51%      | 14.82%     | -23.97% | 0.94   |
| NASDAQ     | 469%         | n.a.       | -33.1%  | n.a.   |

## Risk-return reading

Random Forest delivers the highest terminal wealth, but with higher volatility and deeper drawdowns. Neural Network gives up some return, but the trajectory is cleaner and closer to a professional risk-adjusted profile.

€7.18m

RF final capital

€6.50m

NN final capital

€4.70m

S&P 500 final capital

## 5. Optimization layer: machine-learning selection + Markowitz allocation

After selecting the Top 20% universe, the project adds a maximum-Sharpe mean-variance allocation to test whether risk-based weighting improves portfolio efficiency.

### Optimized portfolios versus benchmarks

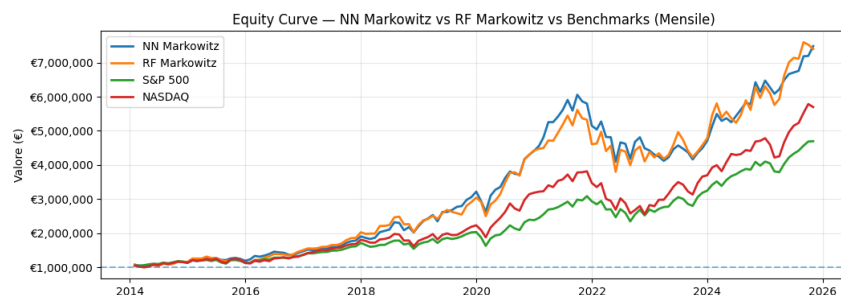


Exhibit: capital evolution of Markowitz strategies

### Risk profile after allocation

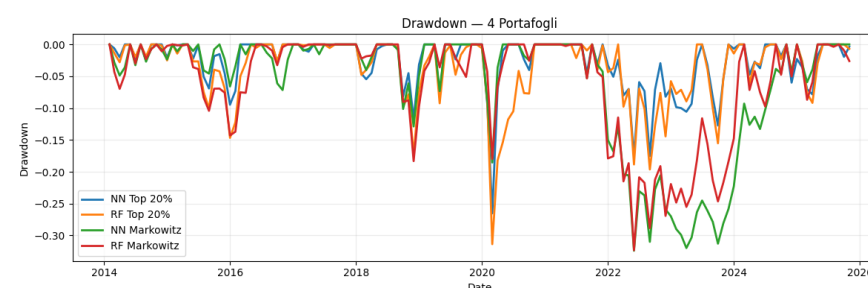


Exhibit: drawdown comparison across portfolios

### Numerical impact of the optimization layer

Monthly metrics for the Top 20% strategies before and after Max-Sharpe allocation

| Portfolio    | Avg monthly return | Monthly vol. | Monthly Sharpe | Max DD  | Total return | Turnover |
|--------------|--------------------|--------------|----------------|---------|--------------|----------|
| NN Top 20%   | 1.464%             | 5.302%       | 0.2761         | -26.53% | 549.57%      | n.a.     |
| NN Markowitz | 1.558%             | 5.140%       | 0.3031         | -32.39% | 648.34%      | 77%      |
| RF Top 20%   | 1.574%             | 5.944%       | 0.2649         | -31.36% | 618.04%      | n.a.     |
| RF Markowitz | 1.586%             | 5.799%       | 0.2736         | -32.32% | 639.96%      | 50%      |

#### Key insight

Optimization improves monthly Sharpe and reduces volatility, but it does not remove drawdown risk. The framework therefore also tracks turnover, transaction costs and scenario losses.

## 6. Robustness: path risk, stress testing and investment conclusions

The final layer evaluates whether the strategies remain credible beyond the historical equity curve by using simulation and crisis-period diagnostics.

### Monte Carlo: NN Markowitz

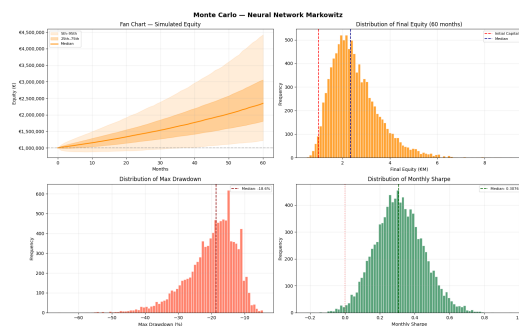


Exhibit: distribution of 10,000 paths over 60 months

### Stress test: 2007-2009

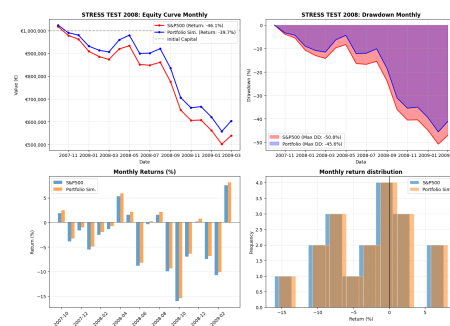


Exhibit: crisis-period portfolio behavior

### Stress test: 2022

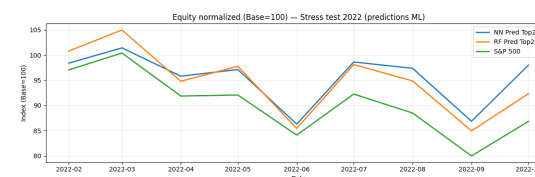


Exhibit: real-prediction stress window

**10,000**

bootstrap simulations

**60m**

forward simulation horizon

**1.6%**

NN probability below €1m

**-7.67%**

NN loss in 2022 stress test

**-13.16%**

S&P 500 loss in 2022 stress test

### Portfolio value proposition

- End-to-end quant workflow: data preparation in R, ML modeling in Python, walk-forward validation, dynamic ranking and rebalanced portfolio construction.
- Professional risk reading: returns are interpreted together with volatility, Sharpe ratio, maximum drawdown, turnover, transaction costs, Monte Carlo dispersion and stress-test losses.
- Practical conclusion: the Neural Network appears more balanced in risk-adjusted terms, while Random Forest is more growth-oriented. Markowitz allocation improves efficiency but does not remove downside risk.

Selected thesis project | Ready for CV, LinkedIn and interview discussion